

Chapter 2

Abstract choice and preference theory

An **economy** is a mathematical object in which agents, resources, firms, etc.(i.e., the elements that constitute the society) are represented.

We start by studying the representation of agents. There are essentially two ways in which an agent is represented in economic models: (i) by means of his/her preferences and (ii) by means of his/her choices. We independently introduce them. We also study the relation between these two approaches.

2.1 Preference theory

Our primitive is a set of alternatives X . In the preference model, an agent is represented by his/her statements of preference about the different alternatives in X . Let us introduce the mathematical structure that allows us to formalize this concept.

2.1.1 Mathematical aside: product space and binary relations

Definition 12. Let A and B be two sets. The **cartesian product of A and B** , denoted $A \times B$, is the set of all lists of two elements in which the first element belongs to A and the second element belongs to B , i.e.,

$$A \times B \equiv \{(a, b) : a \in A, b \in B\}.$$

Example 7. If $A \equiv \{1, 2\}$ and $B \equiv \{1, 2, 5\}$, then

$$A \times B \equiv \{(1, 1), (1, 2), (1, 5), (2, 1), (2, 2), (2, 5)\}.$$

Observe that $(1, 2) \neq (2, 1)$.

2.1.2 Binary relations

A binary relation is a mathematical object that represents statements between pairs of elements of a set. For instance, consider a set of alternatives A . For each two alternatives, say a and b , we collect the information whether an agent, say i , finds a at least as good as b . We can then define the “at least as good” binary relation as the collection of the positive answers.

$$\{(a, b) \in A \times A : i \text{ finds } a \text{ at least as good as } b\}$$

Definition 13. A **binary relation** on a set A is a subset of the cross product space $A \times A$.

In general, this formality is avoided by using short notation that represents the relation that holds between elements. There is no universal notation, however. In economics, there is even multiple notations for the binary relations that represent the statements “agent i finds a at least as good as b .” Some of these are:

- $a R_i b$
- $a \succsim_i b$
- $a \geq_i b$
- $a \succeq_i b$

To each binary relation one can associate its strict and symmetric binary relations.

Definition 14. Let $\succsim$ be a binary relation on a set A .

- The **strict binary relation associated with** $\succsim$, denoted $\succ$, is defined as follows: for each pair $\{a, b\} \subseteq A$, $a \succ b$ if $a \succsim b$ and it is not true that $b \succsim a$.
- The **symmetric binary relation associated with** $\succsim$, denoted $\sim$, is defined as follows: for each pair $\{a, b\} \subseteq A$, $a \sim b$ if $a \succsim b$ and $b \succsim a$.

We refer to the strict relation associated to the “at least as good as” relation as “preference,” as in “agent i prefers a to b ,” we refer to the symmetric relation associated to the “at least as good as” relation as “indifference,”

as in “agent i is indifferent between a and b .” Likewise the different notations for “at least as good” have their own notation for preference and indifference. They are:

at least as good	preference	indifference
$a R_i b$	$a P_i b$	$a I_i b$
$a \succeq_i b$	$a \succ_i b$	$a \sim_i b$
$a \geq_i b$	$a >_i b$	$a \sim_i b$
$a \succeq_i b$	$a \succ_i b$	$a \sim_i b$

We usually assume in economics that binary relations that represent agent’s rankings of alternatives satisfy two consistency properties: completeness and transitivity.

Definition 15. A binary relation $\succeq$ on X is **complete** if for each pair $\{x, y\} \subseteq X$, at least one of the following two statements is true: (i) $x \succeq y$, or (ii) $y \succeq x$.

Definition 16. A binary relation $\succeq$ on X is **transitive** if for each triple $\{x, y, z\} \subseteq X$, if $x \succeq y$ and $y \succeq z$, then $x \succeq z$.

Definition 17. A **rational preference relation** (preference for short) is a complete and transitive binary relation.

In the next section, we will see that when the set on which agents have preferences is finite one can think of a preference as a ranking of the alternatives.

$$\begin{array}{c} R_i \\ \hline a \\ bc \\ d \\ xyz \\ l \end{array}$$

For instance in the preference above: $b I_i c$ and $d P_i l$.

2.2 Utility theory

Is it possible to condense all the information contained in a preference relation in a *welfare index*? The study of this question is known as utility theory.

2.2.1 Mathematical aside: functions

A function from a set A to a set B is a mathematical object that attaches to each element in A a unique element in B . The domain of a function from A to B is the set A and its co-domain B . A function from A to B is generically denoted f . To make explicit the domain and codomain of function f we usually write $f : A \rightarrow B$. The element attached by f to $x \in A$ is denoted $f(x)$; we call $f(x)$ the image of x under f , or the value of f at x . We also say that f maps x to $f(x)$. A statement completely defining a function, shows its name, domain, co-domain, and the image of each element in its domain, for instance,

$$\begin{aligned} f : \quad \mathbb{R}^2 &\rightarrow \mathbb{R} \\ (x_1, x_2) &\mapsto f(x) \equiv x_1 + x_2. \end{aligned}$$

Definition 18. Let A, B, C be three sets. Let $g : A \rightarrow B$ and $f : B \rightarrow C$ be two functions. The **composite function of f and g** , denoted $f \circ g$, is the function:

$$\begin{aligned} f \circ g : \quad A &\rightarrow C \\ x &\mapsto f \circ g(x) \equiv f(g(x)). \end{aligned}$$

2.2.2 Representation of preferences

Definition 19. Let $\succsim$ be a binary relation on X . A function $u : X \rightarrow \mathbb{R}$ **represents** $\succsim$ if for each pair $\{x, y\} \subseteq X$, $x \succsim y$ if and only if $u(x) \geq u(y)$.

Proposition 1. Let $\succsim$ be a binary relation on X . If there is a function $u : X \rightarrow \mathbb{R}$ that represents $\succsim$, then $\succsim$ is complete and transitive, i.e., it is a preference relation.

Proof. See Problem 10. □

A property of a real-valued function is said to be **ordinal** if it is preserved under strictly increasing transformations.

Definition 20. Let $u : X \rightarrow \mathbb{R}$. The **binary relation associated with u** , denoted $\succsim^u$, is defined by: for each pair $\{x, y\} \subseteq X$, $x \succsim^u y$ if and only if $u(x) \geq u(y)$.

The following proposition states that the preference relation associated with a real-valued function is an ordinal property of the function.

Proposition 2. Let $\succsim$ be a binary relation on X that is represented by a function $u : X \rightarrow \mathbb{R}$. For each increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$, we have that $f \circ u$ represents $\succsim$.

Proof. Problem 11, Section 2.2.3. $\square$

The **cardinality** of a set A , denoted $\#A$ or $|A|$, is the number of elements contained in A . Set A is **finite** if $|A| < \infty$.

Theorem 3. Assume X is finite. Let $\succsim$ be a binary relation on X . If $\succsim$ is complete and transitive, then there exists a function $u : X \rightarrow \mathbb{R}$ that represents $\succsim$.

Proof. Let u be the function defined by:

$$\begin{aligned} u : X &\rightarrow \mathbb{R} \\ x &\mapsto u(x) \equiv |\{z : x \succsim z\}|. \end{aligned}$$

We prove that u represents $\succsim$, i.e., for each pair $\{x, y\} \subseteq X$, $x \succsim y$ if and only if $u(x) \geq u(y)$.

1. ($\Rightarrow$) Let $\{x, y\} \subseteq X$ be such that $x \succsim y$. We prove that $u(x) \geq u(y)$.

Let $A \equiv \{z : x \succsim z\}$ and $B \equiv \{z : y \succsim z\}$. We claim that $B \subseteq A$. Let $z \in B$. Then $y \succsim z$. Since $x \succsim y$ and $y \succsim z$, then by transitivity of $\succsim$, $x \succsim z$. Thus, $z \in A$. Since $B \subseteq A$, then $|B| \leq |A|$. Since $u(x) = |A|$ and $u(y) = |B|$, then $u(y) \leq u(x)$.

2. ($\Leftarrow$) Let $\{x, y\} \subseteq X$ be such that $u(x) \geq u(y)$. We prove that $x \succsim y$.

Suppose by means of contradiction that $\neg x \succsim y$. Let $A \equiv \{z : x \succsim z\}$ and $B \equiv \{z : y \succsim z\}$. We claim that $A \subsetneq B$. We prove that $A \subseteq B$ and $B \setminus A \neq \emptyset$. Since $\neg x \succsim y$, then by completeness of $\succsim$, $y \succ x$. First we prove that $A \subseteq B$. Let $z \in A$. Thus, $x \succsim z$. Since $y \succ x$ and $x \succsim z$, then by transitivity of $\succsim$, $y \succ z$. Thus, $z \in B$. Finally, we claim that $y \in B \setminus A$. By completeness of $\succsim$, $y \succsim y$. Thus, $y \in B$. Now, since $\neg x \succsim y$, then $y \notin A$. Thus, $y \in B \setminus A$.

$\square$

The theorem above generalizes to countably infinite sets of alternatives (our proof does not generalize, but an alternative proof does; see Problem 13, Section 2.2.3). It is not true that for an arbitrary set of alternatives an arbitrary preference has a utility representation. We will come back to this problem after introducing more structure in our model.

2.2.3 Problems

1. Let $A \equiv]1, 2] \cup [3, 4]$ and $B \equiv]1, 2] \cup [3, 4]$.¹ Draw $A \times B$ as a subset of $\mathbb{R}^2$. Let $\succsim \equiv A \times B$ be a binary relation on $\mathbb{R}$. Is $\succsim$ symmetric, i.e., for each $\{x, y\} \subseteq X$, $x \succsim y \Rightarrow y \succsim x$? Is it transitive?
2. Let $X \equiv \mathbb{N}$ ($\mathbb{N} \equiv \{1, 2, 3, \dots\}$ is the set of natural numbers). Let $\succsim$ be the binary relation defined by: for each $\{x, y\} \subseteq X$, $x \succsim y$ if and only if $x \leq y$. Is $\succsim$ rational? Describe $\succ$ and $\sim$ as subsets of $\mathbb{N}^2$.
3. Let $\succsim$ be a preference on a set X . See [Kreps \(1988\)](#) for the definition of the properties in the following questions.
 - (a) Prove that $\succ$ is irreflexive, asymmetric, and negatively transitive.
 - (b) Prove that $\sim$ is reflexive, symmetric, and transitive. Is $\sim$ negatively transitive?
4. Ann's preferences on money (i.e., the set of alternatives $X \equiv \mathbb{R}$) are represented by the function $x \in \mathbb{R} \mapsto u_A(x) \equiv x^3$ and Bob's preferences are represented by the function $x \in \mathbb{R} \mapsto u_B(x) \equiv x$. Suppose that both Ann and Bob lose \$100. Could we say that Ann suffers more than Bob from the loss?
5. Let $X \equiv \{1, 2, 3, 4, 5\}$. Construct a binary relation on X that is complete but not transitive. Construct a binary relation on X that is transitive but not complete. Show that both binary relations cannot be represented by a utility function.
6. Let $N \equiv \{1, 2, 3\}$ be a set of agents and $\succsim_1, \succsim_2$, and $\succsim_3$ be three preference relations on the set $A \equiv \{a, b, c\}$. The **majority binary relation induced by $\succsim_1, \succsim_2$, and $\succsim_3$** , denoted M is defined as follows: for each $(x, y) \in A \times A$, $(x, y) \in M$ if and only if $|\{i \in N : x \succsim_i y\}| \geq 2$. Show that M is always complete, but may not be transitive (here you have to construct $\succsim_1, \succsim_2$, and $\succsim_3$ such that the induced M is not transitive.)
7. Let X be a finite set and B a binary relation on X . The **transitive closure** of B is the binary relation on X , denoted T_B , defined as follows: for each pair $\{x, y\} \subseteq X$, $x T_B y$ if and only if there is a list of elements of X , $\{x_i\}_{i=1}^n$ with $n \geq 1$ such that $x_1 = x$, $x_n = y$, $x_1 B x_2$, $x_2 B x_3, \dots$, $x_{n-2} B x_{n-1}$, and $x_{n-1} B x_n$.

¹We denote $]1, 2]$ the set $\{x \in \mathbb{R} : 1 < x \leq 2\}$.

- (a) Prove that $B \subseteq T_B$.
 - (b) State the definition of transitivity of T_B . Prove that T_B is transitive.
 - (c) Let C be a transitive binary relation on X that contains B . Prove that $T_B \subseteq C$.
8. Let $Z \equiv \{z_1, \dots, z_n\}$ be a finite set and X be the set of all non-empty subsets of Z (the set of subsets of Z is called the power set of Z and denoted in mathematics literature 2^Z ; using this notation $X \equiv 2^Z \setminus \{\emptyset\}$; how many elements are in X ?). We say that a preference relation on X is monotone if for each two elements of X , A and B , such that $A \subseteq B$, B is at least as good as A .
- Construct a monotonic preference on X . Provide an economic interpretation of this property when Z is the set of items that may appear in the menu of a restaurant and X is the set of possible menus with items in Z . When would it be an appropriate assumption on agents' preferences?
 - Describe a situation in which monotonicity may be violated. Could a preference on X violate monotonicity?
9. Consider a set $Z \equiv \{a, b, c\}$. The set of lotteries over Z is $\Delta(Z) \equiv \{\pi \equiv (\pi_a, \pi_b, \pi_c) \in \mathbb{R}_+^3 : \pi_a + \pi_b + \pi_c = 1\}$. An Expected Utility Function on $\Delta(Z)$ is a function $U : \Delta(Z) \rightarrow \mathbb{R}$ for which there is $u : Z \rightarrow \mathbb{R}$ such that for each $\pi \in \Delta(Z)$:

$$U(\pi) \equiv \pi_a u(a) + \pi_b u(b) + \pi_c u(c),$$

Prove that a preference $\succsim$ on $\Delta(Z)$ is represented by an Expected Utility Function if and only if there is $v : Z \rightarrow \mathbb{R}_{++}$ such that $\succsim$ is represented by the function that associates with each $\pi \in \Delta(Z)$ the value

$$\Psi(\pi) \equiv v(a)^{\pi(a)} \times v(b)^{\pi(b)} \times v(c)^{\pi(c)}.$$

- 10. Prove Proposition 1.
- 11. Prove Proposition 2.
- 12. Complete the arguments left in the proof of the following theorem.

Theorem 4. Let X be a finite set and $\succsim$ a preference on X . Then there is a function $u : X \rightarrow \mathbb{R}$ that represents $\succsim$.

Proof. The proof proceeds by induction on the cardinality of X .

- (a) Prove that the theorem holds when $|X| = 1$.
- (b) Inductive step. Let $n \in \mathbb{N}$. Suppose that the theorem holds when $|X| \leq n$. Assume that $|X| = n + 1$. We prove that there exists a utility function that represents $\succsim$.
 - i. Prove that there exists $x \in X$ such that for each $y \in X$, $x \succsim y$. (hint: use induction hypothesis and the fact that each finite set of real numbers has a maximum).
 - ii. Fix $x \in X$ such that for each $y \in X$, $x \succsim y$. Such an element of X exists from the claim above. Prove that there is a function $v : X \setminus \{x\} \rightarrow \mathbb{R}$ such that for each $\{y, z\} \subseteq X \setminus \{x\}$, $y \succsim z$ if and only if $v(y) \geq v(z)$. (hint: use induction hypothesis).
 - iii. Identify a candidate u for a representation of $\succsim$ and prove that it is in fact a representation of $\succsim$.

□

13. Let $Z \equiv \{z_1, \dots, z_n\}$ be a finite set of alternatives. Let $\succsim$ be a complete and transitive binary relation on Z . Prove that $\succsim$ is represented by the following utility function: for each $i \in \{1, \dots, n\}$,

$$u(z_i) \equiv \sum_{k: z_i \succsim z_k} \frac{1}{2^k} - \sum_{k: z_k \succ z_i} \frac{1}{2^k}.$$

14. Let X be a finite set and B a binary relation on X . Show that if B is complete and transitive, then there is $x^* \in X$ such that for each $y \in X$, $x^* B y$.
15. Let X be a non-empty finite set and B a binary relation on X . Assume that B satisfies the following two properties:

- (a) If $x B y$ then $\neg(y B x)$ (i.e., it is not true that $y B x$).
- (b) If $\neg(x B y)$ and $\neg(y B z)$, then $\neg(x B z)$.

Let W be the binary relation on X defined by: $x W y$ if $\neg(y B x)$.

- (a) State the definition of completeness of W .
- (b) Prove that W is complete.

- (c) State the definition of transitivity of W .
- (d) Prove that W is transitive.
- (e) Prove that there is an element $x^* \in X$ such that for each $y \in X$, $\neg(y B x^*)$.

2.3 Choice theory

In this section I introduce choice and revealed preference theory. In the **choice model** an agent is represented by his/her choices in the possible choice problems that he/she may face. Our primitive in the choice model is also a set of alternatives X .

2.3.1 Mathematical aside: power set and correspondences

Definition 21. Let A be a set. The **power set of A** , denoted 2^A , is the set of all subsets of A . For each element $B \in 2^A$, $B \subseteq A$. Using this notation, the set of non-empty sets of A is $2^A \setminus \{\emptyset\}$.

Remark 1. Kreps, 1988, denotes $2^X \setminus \{\emptyset\}$ by $P(X)$

Example 8. Let $X \equiv \{1, 2\}$. Then $2^X = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$ and $2^X \setminus \{\emptyset\} = \{\{1\}, \{2\}, \{1, 2\}\}$.

Definition 22. A **correspondence**, C , from a set A to a set B is a mathematical object that associates with each $a \in A$ a subset of B , denoted $C(a) \subseteq B$. The generic correspondence from A to B is denoted $C : A \rightrightarrows B$. Correspondence C is **non-empty valued** if for each $a \in A$, $C(a) \neq \emptyset$.

Technically, a correspondence $C : A \rightrightarrows B$ is a function $C : A \rightarrow 2^B$. An alternative notation for $C : A \rightrightarrows B$ is $C : A \rightsquigarrow B$. The definition of a correspondence should specify its name, domain, co-domain, and the image of each element in its domain. For instance,

$$\begin{aligned} C : \mathbb{N} &\rightrightarrows [0, 1] \\ n &\mapsto C(n) \equiv [0, \frac{1}{n}] \end{aligned}$$

2.3.2 Choice structure

Definition 23. A **choice structure** is a pair $(\mathcal{B}, C)$ where $\mathcal{B}$ is a non-empty family of non-empty subsets of X (i.e., $\mathcal{B} \subseteq 2^X \setminus \{\emptyset\}$) and $C : \mathcal{B} \rightrightarrows X$ is a non-empty valued correspondence such that for each $B \in \mathcal{B}$, $C(B) \subseteq B$.

We interpret $\mathcal{B}$ as a family of possible choice problems that the agent may face. For each choice problem $B \in \mathcal{B}$, the agent selects $C(B)$ as his/her best choices.

In the same spirit of rationality of preferences, one may be interested in studying choice structures that satisfy certain consistency properties. The following property has played a central role in the literature.

Intuition: Let $(\mathcal{B}, C)$ be a choice structure, $A \in \mathcal{B}$ a choice problem and $x \in A$. Suppose that $x \in C(A)$. This reveals that the agent favors the choice of x over all the other elements of A . Let $y \in A$ and suppose that in another choice problem, say B , both x and y are available and y is chosen at B . It is “reasonable” to think that the agent will chose also x at B , since at choice problem A the agent revealed being in favor of x over y . If y makes it to the top, so does x .

Definition 24. A choice structure $(\mathcal{B}, C)$ satisfies the **Weak Axiom of Revealed Preference (WARP)** if for each two elements of $\mathcal{B}$, A and B , and each pair $\{x, y\} \subseteq A \cap B$, if $x \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

Remark 2. WARP is also known in the literature as Houthakker’s axiom, c.f., [Kreps \(1988\)](#).

When the family of choice problems $\mathcal{B}$ is rich enough, WARP may be divided into two independent simpler properties.

The first property is interpreted as: “The champion of the world is the champion of his/her own country.”

Definition 25. A choice structure $(\mathcal{B}, C)$ satisfies **Sen’s property α** if for each two elements of $\mathcal{B}$, A and B such that $A \subseteq B$, and each $x \in C(B) \cap A$, we have that $x \in C(A)$.

The second property is interpreted as: “If the champion of a country is the champion of the world, then all champions of that county are champions of the world.”

Definition 26. A choice structure $(\mathcal{B}, C)$ satisfies **Sen’s property β** if for each two elements of $\mathcal{B}$, A and B such that $A \subseteq B$, and each $\{x, y\} \subseteq C(A)$, we have that $x \in C(B)$ implies $y \in C(B)$.

Proposition 3. Let $(\mathcal{B}, C)$ be a choice structure such that for each $B \in \mathcal{B}$ and each $\emptyset \neq B' \subseteq B$, $B' \subseteq B$. Then, $(\mathcal{B}, C)$ satisfies WARP if and only if it satisfies both Sen’s properties α and β .

Proof. Let $(\mathcal{B}, C)$ be as in the statement of the proposition.

($\Rightarrow$) Suppose that $(\mathcal{B}, C)$ satisfies WARP. We prove it satisfies Sen’s α and β .

- Property α . Suppose that $A \subseteq B \subseteq X$ and $x \in C(B) \cap A$. We want to prove $x \in C(A)$. Since C is non-empty valued, then $C(A) \neq \emptyset$. Let $y \in C(A)$. Since $A \subseteq B$, then $y \in B$. Since $(\mathcal{B}, C)$ satisfies WARP then $\{x, y\} \subseteq A \cap B$, $x \in C(B)$ and $y \in C(A)$ implies $x \in C(A)$.

- Property β . Suppose that $A \subseteq B \subseteq X$, $\{x, y\} \subseteq C(A)$, and $x \in C(B)$. We want to prove that $y \in C(B)$. Since $C(A) \subseteq A \subseteq B$ then $\{x, y\} \subseteq A \cap B$. Since $(\mathcal{B}, C)$ satisfies WARP then $\{x, y\} \subseteq A \cap B$, $x \in C(B)$ and $y \in C(A)$ imply $y \in C(B)$.

($\Leftarrow$) Suppose that $(\mathcal{B}, C)$ satisfies Sen's α and β . We prove that it satisfies WARP. Let A and B be two non-empty subsets of X and $\{x, y\} \subseteq A \cap B$ be such that $x \in C(A)$ and $y \in C(B)$. We want to prove that $x \in C(B)$. Let $D \equiv A \cap B$. Since $\{x, y\} \subseteq A \cap B$, then $D \neq \emptyset$. By hypothesis, $D \in \mathcal{B}$. Thus, $\emptyset \neq C(D) \subseteq D$. Now, since $(\mathcal{B}, C)$ satisfies Sen's α , then $D \subseteq B$ and $y \in C(B) \cap D$ imply that $y \in C(D)$. By the same argument, $D \subseteq A$ and $x \in C(A) \cap D$ imply that $x \in C(D)$. Since $(\mathcal{B}, C)$ satisfies Sen's β , then $D \subseteq B$, $\{x, y\} \subseteq C(D)$, and $y \in C(B)$ imply $x \in C(B)$. $\square$

2.3.3 Revealed preference theory

Is there a connection between the preference model and the choice model? Revealed preference theory answers this question.

First, let us consider the choices made by an agent represented by a binary relation $\succsim$ (observe that I did not require the agent to be “rational.”)

Definition 27. Let $\succsim$ be a binary relation on X . For each non-empty set $B \subseteq X$, let $C^{\succsim}(B)$ be the set

$$C^{\succsim}(B) \equiv \{x \in B : \text{for each } y \in B, x \succsim y\}.$$

Let $\mathcal{B} \subseteq 2^X \setminus \{\emptyset\}$. The following proposition states that when X is finite and $\succsim$ is a preference relation, $(\mathcal{B}, C^{\succsim})$ is a choice structure.

Proposition 4. Let $\succsim$ be a preference relation on a finite set X . Then for each non-empty $B \subseteq X$, $C^{\succsim}(B) \neq \emptyset$.

Proof. Problems 2–5, Section 2.3.4. $\square$

There are conditions, besides finiteness of choice problems, that guarantee the choice correspondence $C^{\succsim}$ associated with a preference relation is non-empty valued. We will assume in what follows that those conditions are met. Thus, for each family of non-empty sets of X , $\mathcal{B}$, and each preference relation $\succsim$, we refer to $(\mathcal{B}, C^{\succsim})$ as a choice structure.

Our main result here is that the choices of an agent represented by a preference relation satisfy WARP.

Proposition 5. Let $\succsim$ be a preference relation on X and $\mathcal{B}$ be a family of non-empty sets of X . Then $(\mathcal{B}, C^{\succsim})$ satisfies WARP.

Proof. Let $\succsim$ be a preference relation on X . We prove that $(\mathcal{B}, C^\succsim)$ satisfies WARP. Let A and B be two non-empty subsets of X and $\{x, y\} \subseteq A \cap B$ be such that $x \in C^\succsim(A)$ and $y \in C^\succsim(B)$. We prove that $x \in C^\succsim(B)$. Since $x \in C^\succsim(A)$ then

$$x \in \{z \in A : \text{for each } w \in A, z \succsim w\}.$$

Since $y \in A$, then $x \succsim y$. Now, since $y \in C^\succsim(B)$ then $y \in \{z \in B : \text{for each } w \in B, z \succsim w\}$. Let $w \in B$. Since $x \succsim y$ and $y \succsim w$, then by transitivity of $\succsim$, $x \succsim w$. Since w is an arbitrary element of B , then $x \in \{z \in B : \text{for each } w \in B, z \succsim w\}$. Thus, $x \in C^\succsim(B)$. $\square$

The symmetric question is equally interesting. Given a choice structure that satisfies WARP, is there a preference that induces these choices?

Definition 28. A preference relation $\succsim$ **rationalizes** a choice structure $(\mathcal{B}, C)$ if for each $B \in \mathcal{B}$, $C(B) = C^\succsim(B)$.

The following example shows that the answer to the above question may be negative.

Example 9. $X \equiv \{x, y, z\}$, $\mathcal{B} \equiv \{\{x, y\}, \{x, z\}, \{y, z\}\}$, $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$, $C(\{x, z\}) = \{z\}$. To prove that $(\mathcal{B}, C)$ satisfies WARP observe that for any two elements of $\mathcal{B}$, say A and B , if $A \neq B$ then there is no pair $a \in C(A)$ and $b \in C(B)$ such that $\{a, b\} \subseteq A \cap B$. So it is trivially satisfied. However, the choice structure cannot be rationalized by a complete and transitive binary relation. Could you see why?

We now develop conditions that guarantee the existence of a preference relation that represents a given choice structure. Let us first consider the following question: given a choice structure $(\mathcal{B}, C)$, what is the preference relation that rationalizes $(\mathcal{B}, C)$, if any?

Definition 29. The **revealed binary relation** associated with a choice structure $(\mathcal{B}, C)$, denoted $\succsim(\mathcal{B}, C)$, is defined by: for each $\{x, y\} \subseteq X$, $x \succsim(\mathcal{B}, C) y$ if and only if: (i) $x = y$, or (ii) there is $B \in \mathcal{B}$ such that $\{x, y\} \subseteq B$ and $x \in C(B)$.

Definition 30. A family of non-empty sets of X , $\mathcal{B}$, satisfies the **pair-and-triple condition** if it contains, at least, all subsets of two and three elements of X .

Proposition 6. If $(\mathcal{B}, C)$ satisfies WARP and $\mathcal{B}$ satisfies the pair-and-triple condition, then $\succsim(\mathcal{B}, C)$ is a preference relation.

Proof. We prove that $\succsim(\mathcal{B}, C)$ is complete and transitive.

First, we prove that $\succsim(\mathcal{B}, C)$ is complete. Let $\{x, y\} \subseteq X$. If $x = y$, then, $x \succsim(\mathcal{B}, C) y$. If $x \neq y$, then since $B \equiv \{x, y\} \in \mathcal{B}$ and C is non-empty valued, then either $x \in C(B)$ or $y \in C(B)$. Thus, either $x \succsim(\mathcal{B}, C) y$ or $y \succsim(\mathcal{B}, C) x$.

Second, we prove that $\succsim(\mathcal{B}, C)$ is transitive. Let $\{x, y, z\} \subseteq X$ be such that $x \succsim(\mathcal{B}, C) y$ and $y \succsim(\mathcal{B}, C) z$. We prove that $x \succsim(\mathcal{B}, C) z$. Let $A \equiv \{x, y, z\}$. Since $\mathcal{B}$ satisfies the pair-and-triple condition, then $B \in \mathcal{B}$. We claim that $x \in C(B)$. Consider the following two cases.

Case 1. $y \in C(B)$: since $x \succsim(\mathcal{B}, C) y$, then there is $A \in \mathcal{B}$ such that $\{x, y\} \subseteq A$ and $x \in C(A)$. Since $\{x, y\} \subseteq A \cap B$, $y \in C(B)$ and $x \in C(A)$, then by WARP, $x \in C(B)$.

Case 2. $z \in C(B)$: since $y \succsim(\mathcal{B}, C) z$, then there is $A \in \mathcal{B}$ such that $\{y, z\} \subseteq A$ and $y \in C(A)$. Since $\{y, z\} \subseteq A \cap B$, $z \in C(B)$ and $y \in C(A)$, then by WARP, $y \in C(B)$. By Case 1, $x \in C(B)$.

Since C is non-empty valued, then either $x \in C(B)$ or at least one of the two cases above holds. Thus, $x \in C(B)$. Since $\{x, z\} \subseteq B$ and $x \in C(B)$, then $x \succsim(\mathcal{B}, C) z$. $\square$

If a choice structure satisfies WARP and the pair-and-triple condition, then we can refer to its revealed binary relation as its **revealed preference relation**.

Theorem 5. If $(\mathcal{B}, C)$ satisfies WARP and $\mathcal{B}$ satisfies the pair-and-triple condition, then $\succsim(\mathcal{B}, C)$ rationalizes $(\mathcal{B}, C)$. Moreover, if $\succsim'$ is a preference relation that rationalizes $(\mathcal{B}, C)$, then $\succsim'$ is equal to $\succsim(\mathcal{B}, C)$.

Proof. Suppose that $(\mathcal{B}, C)$ satisfies WARP and $\mathcal{B}$ satisfies the pair-and-triple condition. By Proposition 6, $\succsim(\mathcal{B}, C)$ is a preference relation. We claim that for each $B \in \mathcal{B}$, $C(B) = C^{\succsim(\mathcal{B}, C)}(B)$. For simplicity we denote $D \equiv C^{\succsim(\mathcal{B}, C)}$.

Let $B \in \mathcal{B}$. We claim that $C(B) = D(B)$. We prove it in two parts.

First, let $x \in C(B)$. We prove that $x \in D(B)$. Let $y \in B$. Since $x \in C(B)$ then $x \in B$. Thus, $\{x, y\} \subseteq B$ and $x \in C(B)$. Thus, $x \succsim(\mathcal{B}, C) y$. Since y is an arbitrary element of B , then $x \in D(B)$.

Second, let $x \in D(B)$. We prove that $x \in C(B)$. Let $y \in C(B)$. Suppose without loss of generality that $y \neq x$. Since $x \in D(B)$, then $x \succsim(\mathcal{B}, C) y$. Thus, there is $A \in \mathcal{B}$ such that $\{x, y\} \subseteq A$ and $x \in C(A)$. Thus, $\{A, B\} \subseteq \mathcal{B}$, $\{x, y\} \subseteq A \cap B$, $x \in C(A)$ and $y \in C(B)$. By WARP, $x \in C(B)$.

Since $C(B) \subseteq D(B)$ and $D(B) \subseteq C(B)$, then $C(B) = D(B)$.

It is left to prove that if $\succsim'$ is a preference relation that rationalizes $(\mathcal{B}, C)$, then $\succsim' = \succsim(\mathcal{B}, C)$. Let $\{x, y\} \subseteq X$. If $x \succsim' y$, then $x \in C(\{x, y\})$. Thus, $x \succsim(\mathcal{B}, C) y$. If $x \not\succsim(\mathcal{B}, C) y$, then there is $B \in \mathcal{B}$ such that $\{x, y\} \subseteq B$ and $x \in C(B)$. By WARP, $x \in C(\{x, y\})$. Since $\succsim'$ rationalizes $(\mathcal{B}, C)$, then $x \succsim' y$. $\square$

2.3.4 Problems

1. Prove Proposition 3.
2. Assume that X is finite and let $\succsim$ be a binary relation on X . For each $B \subseteq X$, let $C^\succ(B)$ be the set

$$C^\succ(B) \equiv \{x \in B : \text{there is no } y \in B, \text{ s.t. } y \succ x\}.$$

Prove that for each binary relation $\succsim$ and each family of non-empty subsets of X , $\mathcal{B}$, $(\mathcal{B}, C^\succ)$ satisfies Sen's property α .

3. Let $\succsim$ be a binary relation on a set X . Prove that if $\succsim$ is complete and transitive, then for each $B \subseteq X$, $B \neq \emptyset$, $C^\succ(B) = C^\sim(B)$. (See Problem 2.)
4. A binary relation $\succsim$ on X is $\succ$ -acyclic if there is no $\{x_1, \dots, x_n\} \subseteq X$ such that for each $i = 1, \dots, n-1$, $x_i \succ x_{i+1}$ and $x_1 = x_n$. Prove that if X is finite, then $\succsim$ is $\succ$ -acyclic if and only if for each $B \subseteq X$, $B \neq \emptyset$, $C^\succ(B) \neq \emptyset$.
5. Let $\succsim$ be a binary relation on a set X . Prove that if $\succsim$ is complete and transitive and $B \neq \emptyset$ is finite, then $C^\sim(B)$ is non-empty. Hint: does rationality of $\succsim$ imply $\succ$ -acyclicity?
6. (Kreps, 1988, problem 9, Pag. 17). Give an example of a finite set X and a choice correspondence C on the set of non-empty sets of X such that the choice structure $(2^X \setminus \{\emptyset\}, C)$ satisfies Sen's property α but cannot be rationalized by a preference relation.
7. Let $L \in \mathbb{N}$. Assume that $X \equiv \mathbb{R}_+^L$. The competitive (Walrasian) budget set at prices $p \in \mathbb{R}_+^L$ and wealth $w \in \mathbb{R}_+$ is the set $B(p, w) \equiv \{x \in X : p \cdot x \leq w\}$. Let $\mathcal{B}^W \equiv \{B(p, w) : p \in \mathbb{R}_+^L, w \in \mathbb{R}_+\}$ be the family of competitive budget sets ($\mathcal{B}^W$ is the family of menus that an agent may face in a competitive environment). Let $C : \mathcal{B}^W \rightrightarrows X$ be a correspondence such that for each $B \in \mathcal{B}^W$, $\emptyset \neq C(B) \subseteq B$.

Correspondence C collects all the information of an agents' market behavior.

Does the family of sets $\mathcal{B}^W$ satisfy the pair-and-triple condition?

Chapter 3

Consumer theory

3.1 Mathematical aside

3.1.1 Euclidean space and Euclidean distance

A **space** is a list $(Y, t_1, \dots, t_n)$ in which the first component is a set and the other components are “tools” that are available to work in the space. A **metric space** is a set in which one can measure distances between elements, i.e., (Y, d) where Y is a set and $d : X \times X \rightarrow \mathbb{R}$ is a function, called distance, that has the following properties: (1) for each $\{x, y\} \subseteq Y$, $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$; (2) $d(x, y) = d(y, x)$; and (3) for each $z \in Y$, $d(x, y) \leq d(x, z) + d(z, y)$.

Let L be a natural number. The set $\mathbb{R}^L$ is the set of lists of L real numbers. For instance $(1, \sqrt{2}, \pi) \in \mathbb{R}^3$. The generic element in $\mathbb{R}^L$ is denoted $x \equiv (x_l)_{l=1}^L$. The set $\mathbb{R}^L$ is known as the L -dimensional Euclidean space.

Definition 31. For each $\{x, y\} \subseteq \mathbb{R}^L$, the Euclidean distance between x and y is:

$$\|x - y\| \equiv \left(\sum_{l=1}^L |x_l - y_l|^2 \right)^{\frac{1}{2}}$$

In what follows $X \subseteq \mathbb{R}^L$.

Theorem 6. Let $d_E : X \times X$ be the function $(x, y) \in X \times X \mapsto d_E(x, y) \equiv \|x - y\|$. Then, the space (X, d_E) is a metric space, i.e., d_E is a distance.

Definition 32. Let $x \in X$ and $\varepsilon \in \mathbb{R}_{++}$. The open ball centered at x with radius ε in X is the set: $B_\varepsilon^X(x) \equiv \{y \in X : d_E(x, y) < \varepsilon\}$.

Definition 33. A set $A \subseteq X$ is **open in X** if for each $x \in A$ there is $\varepsilon \in \mathbb{R}_{++}$ such that $B_\varepsilon^X(x) \subseteq A$.

Definition 34. A set $A \subseteq X$ is **closed in X** if its complement in X is open, i.e., if $X \setminus A$ is open in X .

Remark 3. A closed set is not necessarily a set that is not open. There are sets that are neither closed, nor open. For instance, $(0, 1]$ is not open or closed in $\mathbb{R}$. There are sets that are closed and open simultaneously. Could you give an example in $\mathbb{R}^L$?

Definition 35. A sequence $\{x^t\}_{t \in \mathbb{N}}$ in X is a function from $\mathbb{N}$ into X . The value of the sequence at $t \in \mathbb{N}$ is $x^t \equiv (x_1^t, \dots, x_L^t) \in X$.

Definition 36. A sequence $\{x^t\}_{t \in \mathbb{N}}$ in X converges to $x \in X$ if for each $\varepsilon \in \mathbb{R}_{++}$ there exists $T(\varepsilon) \in \mathbb{N}$ such that for each $t \in \mathbb{N}$, $t \geq T(\varepsilon)$, $d(x^t, x) < \varepsilon$. If $\{x^t\}_{t \in \mathbb{N}}$ converges to x we write $x^t \xrightarrow[t \rightarrow \infty]{} x$ and define $\lim_{t \rightarrow \infty} x^t \equiv x$.

Lemma 1. A set $G \subseteq X$ is closed in X if and only if for each convergent sequence $x^t \xrightarrow[t \rightarrow \infty]{} x$, such that for Each $t \in \mathbb{N}$, $x^t \in G$, we have that $x \in G$.

Definition 37. A set $H \subseteq X \times X$ is closed in $X \times X$ if for any two convergent sequences in X , $x^t \xrightarrow[t \rightarrow \infty]{} x$ and $y^t \xrightarrow[t \rightarrow \infty]{} y$ such that for each $t \in \mathbb{N}$, $(x^t, y^t) \in H$, we have that $(x, y) \in H$.

3.1.2 Inner product and half spaces

Definition 38. Let $x \equiv (x_l)_{l=1}^L \in \mathbb{R}^L$ and $y \equiv (y_l)_{l=1}^L \in \mathbb{R}^L$. The inner product between x and y is defined by:

$$x \cdot y \equiv x_1 y_1 + \dots + x_L y_L.$$

Definition 39. Let $p \equiv (p_l)_{l=1}^L \in \mathbb{R}^L$ and $w \in \mathbb{R}$. The hyperplane associated with p and w is the set:

$$h(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x = w\}.$$

Each hyperplane determines two closed half spaces and two open half spaces:

Definition 40. Let $p \equiv (p_l)_{l=1}^L \in \mathbb{R}^L$ and $w \in \mathbb{R}$.

- The positive closed half-space associated with p and w is the set:

$$h^{\geq}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x \geq w\}.$$

- The negative closed half-space associated with p and w is the set:

$$h^{\leq}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x \leq w\}.$$

- The positive open half-space associated with p and w is the set:

$$h^{>}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x > w\}.$$

- The negative open half-space associated with p and w is the set:

$$h^{<}(p, w) \equiv \{x \in \mathbb{R}^L : p \cdot x < w\}.$$

Geometrically, p is the vector normal to $h(p, w)$. Let $x \in h(p, w)$. If $x' \in h^{>}(p, w)$ then $p \cdot (x' - x) > 0$. If $x' \in h(p, w)$, then $p \cdot (x' - x) = 0$. If $x' \in h^{<}(p, w)$, then $p \cdot (x' - x) < 0$.

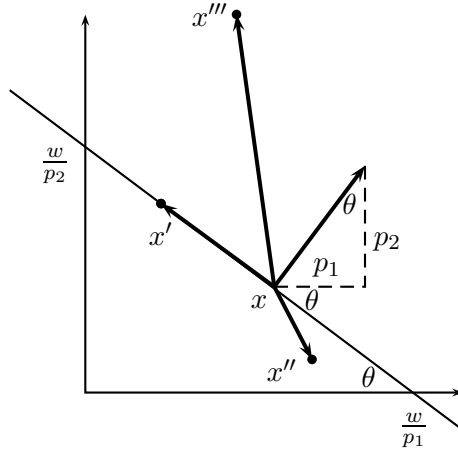


Figure 3.1: half spaces.